The Repair Saturates the Gate: Context Noise as a Degeneracy Diagnostic in Active-Dendrite Continual Learning

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Abstract

Active-dendrite continual-learning systems condition each neuron's plasticity and routing on a task-context signal. A companion audit ([1], S1) found that such networks never develop segment–task specialization and that a constant bias b on the own-segment dendritic input causally repairs the learning term ($\Delta\ell$ up to +4.6, $|t| \geq 8$). Every result in that line – including the repair itself – was obtained under oracle context. We ask the question the line left open: *when the context signal is wrong, what survives?* Under the worst-case evaluation protocol (each sample's context one-hot flipped to a uniformly random other task with probability ρ), the answer is: everything, at every noise level tested. Final accuracy moves by at most 0.25 points at $\rho \leq 0.4$ for the base system and by at most 0.01 points for the repaired one; the repair's paired gain (+4.80, t=17.0, on MNIST) is retained at 100–105% under noise on MNIST and 100–124% on Fashion-MNIST. Zero noise sensitivity is not robustness — it is a symptom. Mechanism probes show why: the trained dendritic gate is *context-degenerate*. Its value under the correct context and under every wrong context is identical to four decimals (0.565); after the repair (b=10) context swaps change at most 0.02% of predictions on average

(worst pair 0.15%); and the dendritic weight columns stay mutually orthogonal (cosine $\approx$ 0.003) throughout training. Two degenerate controls then locate what the repair actually repairs. A *constant-gate* control that pins the gate at its measured baseline value (0.565) reproduces the base network, not the repair — merely holding the gate constant buys nothing, because the base system already runs at that constant. A *no-gate* control ($h = \mathrm{kwta}(t)$, same trunk, same anchor chain) recovers most of the repair's benefit on MNIST (67.50 vs. 68.67, -1.16, $t = -2.7$) and numerically exceeds it on Fashion-MNIST (44.27 vs. 42.89, n.s.): the repair functions as a saturating soft deletion of the gate ($\sigma(10+u) \approx 1$), and what it buys is the removal of a constant $\approx$0.57 damping factor — not restored routing. We conclude that "context-gated protection" in this architecture family runs, at these training budgets, on a gate that does not gate by context; noise-robustness certificates for such systems should therefore be read as *degeneracy diagnostics*, and repairs that survive adversarial context corruption deserve particular scrutiny.

Keywords: continual learning, active dendrites, context gating, robustness evaluation, measurement validity, catastrophic forgetting

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1. Introduction

Active-dendrite architectures give each neuron a bank of dendritic segments and let a task-context signal select which segment gates the neuron's output [2, 3, 4]. In continual learning the arrangement is attractive for a specific reason: if each task's context activates its own segments, protection can be localized to exactly the parameters that task used, and interference

across tasks should shrink. A sequence of audits from the same laboratory has stress-tested that promise. The original DeLP audit[1] corrected its own baselines and found most of the reported advantage was calibration, not mechanism. The companion dose–response audit (S1, [1]) found the specialization premise itself fails: segment–task routing never specializes, the standard routing instrument is contaminated by dead neurons, and yet a one-parameter intervention — a constant bias b on the own-segment dendritic input — causally improves the learning term in every cell of a $\{\text{dataset}\} \times \{\text{depth}\} \times \{\text{granularity}\}$ grid. A transfer study (S3, [6]) then showed the repair does not survive stronger backbones. One assumption has been shared by every experiment in this line, and by most of the active-dendrite continual-learning literature: *the context signal at evaluation is the correct one.*

This paper removes that assumption and reports what happens. The probe is deliberately adversarial: at evaluation time, each sample's context one-hot is flipped to a uniformly random *other* task with probability $\rho \in \{0.1, 0.2, 0.4\}$. Training is untouched. Three questions, preregistered before any data was collected (Sec. 3): (Q1) does the base gated system degrade? (Q2) does the S1 repair — whose premise is precisely that the gate should route by context — survive context corruption? (Q3) does a soft mixture mitigation help?

The empirical answers are monotone: nothing degrades, at any ρ, on either dataset, with or without the repair (Sec. 4). A robustness result this

[1] "Where do the gains come from?", [5]; the precursor study.

clean is not a reason to celebrate; it is a reason to dig. The rest of the paper is the dig. Mechanism probes (Sec. 5) show the trained gate is **context-degenerate**: its value is identical to four decimals under every context, and after the repair the network's predictions are invariant to context swaps up to 0.02% of samples. Two degenerate controls (Sec. 6) then locate what the repair actually does: pinning the gate at its own baseline constant reproduces the base network — flattening buys nothing, because the base system already runs at that constant — while deleting the gate outright recovers most of the repair's benefit on MNIST and numerically all of it on Fashion-MNIST. The repair works by *saturating the gate to a constant gain of one*: a soft deletion of the system's dependence on the component it was supposed to be fixing.

The contribution is threefold:

1. **A negative-space result with a warning attached.** Adversarial context corruption is free at every noise level — because the mechanism it corrupts is not load-bearing at these training budgets. Any "robust to context noise" certificate earned by such a system is evidence of degeneracy, not of robustness.
2. **A mechanism re-qualification of the S1 repair.** The own-segment bias does not restore segment–task routing (which S1 showed never existed); it saturates the gate into a context-free gain of one — a soft deletion of the dendritic damping. A constant-gain control at the measured baseline gate value reproduces the base system, and a gate-deletion control reproduces most (MNIST) or more (Fashion-MNIST) of the repair: what helps is *removing* the constant ≈ 0.57 damping, not shaping it.
3. **A reusable three-instrument protocol**: worst-case noise sweep →

flip-rate/gate-response probe → degenerate-control ablation with two variants (constant-gate and gate-removal). Each instrument is cheap; together they convert a suspicious null into a mechanistic verdict. We preregistered the first two, and the controls were forced on us by their result.

2. Background and setting

Architecture.. The reference network is the one shared by the S1/S3 line [1, 6]: a one-hidden-layer active-dendrite network (ADNet) with n_h=256 hidden units, one dendritic segment bank per task (K=T), k-winners-take-all sparse output (k=64). Given input x and context one-hot c:

$$t = W_{\text{ff}}x, \qquad u_k = w_k^\top c \;\; (k\text{-th segment row}), \qquad h = \text{kwta}(t \odot \sigma(d)), \quad d = \max_k u_k, \tag{1}$$

i.e. the winning segment's scalar d gates every hidden unit multiplicatively. The repair of S1 adds a constant b to the own segment's u_{own} before the max, everywhere (train, eval, usage) [1].

Protection.. Usage-localized elastic consolidation (DeLP [5]): after each task, a Fisher anchor is stored and weighted by measured per-neuron usage; the penalty pulls shared parameters toward anchored values with strength λ=1000 (matching S1's dose–response setting). Segment-specific gradients are protected so each task trains its own dendritic block.

Protocol.. Permuted-MNIST and Permuted-Fashion-MNIST, T=10 tasks, 10 seeds, paired across all arms; permutations drawn once per seed and shared.

Metrics: final average accuracy $\bar{a}$ over all tasks after the last task; paired t-statistics over seeds.

Execution correction (2026-10-27).. An independent data-consistency audit of this line, run after the first version of these experiments had completed, found two defects in the harness and the experiments were re-run in full. (i) The elastic penalty was applied without its λ factor, so every run had effectively executed with $\lambda \approx 1$ while the record claimed λ=1000; the penalty now carries λ as the reference implementation does. (ii) The degenerate control had a hard-coded hidden width of 2000 while the dendritic arms use 256, so the two arms of the ablation differed by roughly eight-fold in parameter count; both now use the same width. The numbers reported here are all from the corrected re-run; the first version's records are retained in the artifact under an explicit deprecation notice. The correction moves the absolute accuracies substantially — a genuinely enforced penalty forgets more, and the base network drops ∼7.6 points on MNIST — and it changes one qualitative verdict: the first version's MNIST gate-removal *equivalence* was an artifact of the width mismatch, and under matched widths the equivalence becomes approximate rather than exact (Sec. 6). The robustness null and the mechanism probes are unaffected.

3. Preregistered design

The design was locked on 2026-10-25, before any data was written to disk (preregistration document included in the artifact). Noise is injected *at evaluation only*; training sees oracle contexts. The noise model is worst-case in a specific sense: with probability ρ per sample the context one-hot

is replaced by a uniformly random *other* task — never the correct one — so corrupted samples exercise the gate under a plausible wrong context rather than a neutral one.

- **E1 (gating robustness).** b=0, $\rho \in \{0, 0.1, 0.2, 0.4\}$, both datasets. Survival: ρ=0.1 costs ≤ 2 points and ρ=0.2 costs ≤ 5, *and* the per-seed direction agrees across both datasets in $\geq 8/10$ seeds; death: ρ=0.1 costs > 5 or ρ=0.2 collapses ($\bar{a} < 50$ or cost > 10).

- **E2 (repair survival).** Same grid crossed with $b \in \{0, 10\}$, paired per $(\rho, \text{dataset}, \text{seed})$. Survival: at ρ=0.1 the repair's paired $\Delta\bar{a} > 0$ and retains $\geq 50\%$ of its ρ=0 gain; death: $\Delta\bar{a} \leq 0$.

- **E3 (soft mixture, exploratory).** Context replaced by $(1-\rho)$ onehot+ ρ uniform; direction only, no survival claim (multiple-comparison hygiene).

Two deviations from the pre-registered wording, both forced and both reported rather than absorbed. First, E2's criterion was written for $\Delta\ell$; the noise harness records only the final-phase per-task accuracies and no ℓ field, so $\Delta\ell$ is not computable from the collected data and the criterion is evaluated on $\Delta\bar{a}$ instead — the same "net benefit of the repair" quantity, in the one form the runs support. Second, E1's third condition (per-seed directional agreement) was not met in strict form: the per-seed ρ-effect has mixed signs, though every per-seed shift stays below 0.1 points. We report that as a condition we did not satisfy rather than dropping it, and the two amplitude conditions pass with orders of magnitude to spare.

Table 1: Final average accuracy under worst-case context flips (n=10 seeds; noise applied to the final-phase evaluation only; λ=1000 enforced, n_h=256 in all arms).

dataset	arm	ρ=0	ρ=0.1	ρ=0.2	ρ=0.4
MNIST	base (b=0)	63.86	63.79	63.72	63.62
MNIST	repaired (b=10)	68.67	68.67	68.67	68.67
Fashion	base (b=0)	42.20	42.15	42.09	42.04
Fashion	repaired (b=10)	42.89	42.89	42.89	42.89

The noise stream is fixed per seed (a dedicated generator), so paired comparisons share corruption patterns exactly. The protocol trains one model per $(\text{dataset}, b, \text{seed})$ and evaluates it at all four ρ levels: **60 training runs** (40 for E1+E2, 20 for E3) yielding 240 evaluation cells. (The pre-registration counted ρ as independent training runs — 240 — which over-counts by a factor of four; the number of evaluated $(\rho, b, \text{dataset}, \text{seed})$ cells is unchanged.) A dated amendment documenting every deviation is included in the artifact ("prereg/S4_ctxnoise_ _ _2026-10-27.md"); the original pre-registration text is left untouched.

4. Results: nothing degrades, at any noise level

Table 1 is the entire empirical content of E1 and E2. Both preregistered survival criteria pass with orders of magnitude to spare: the largest movement anywhere is 0.25 points (MNIST base, ρ=0.4), the largest on Fashion-MNIST is 0.16, and the repaired arm moves by at most 0.01 points on MNIST and *exactly* 0.000 points on Fashion-MNIST — where the death criteria allowed 10 and 5. The repair's paired gain on MNIST is +4.80 (t=17.0) at ρ=0 and

+5.05 (t=17.3) at ρ=0.4; retention of the gain is 100.0% → 105.1% across the sweep — under noise the repair *helps slightly more*, because its floor is flatter. On Fashion-MNIST the gain is small, as S3 found at T=10 [6]: +0.69 (t=1.5, n.s.) at ρ=0, rising to +0.86 (t=1.8) at ρ=0.4 (retention 100% → 123.7%; all n.s.). E1's third condition (per-seed directional agreement) holds in strict form on MNIST — 10/10 seeds shift negatively at every ρ — and fails on Fashion-MNIST (8/10, 9/10, 7/10 negative, with every per-seed shift within ±0.18 points); we report it as satisfied on one dataset and not on the other, as disclosed in Sec. 3. E3's soft mixture tracks the oracle slightly more closely than hard flips on the base arm (MNIST ρ=0.4: soft −0.09 vs. hard −0.25; paired soft−hard +0.16, t=3.8) and does nothing on the repaired arm; every E3 effect is ≤0.26 points.

A protocol-level artifact is worth excluding explicitly: the noise *is* reaching the network. Per-seed corruption patterns are fixed, the context tensor is modified before the forward pass, and — the direct check — swapping contexts *does* change predictions in the base network (14.8% of samples on average on MNIST, 30.1% on Fashion-MNIST, up to 51.7% for particular swaps; Sec. 5). What is true is that prediction changes of this kind average out across a 10-way task mixture at the final-accuracy level. The null is real, and it demands a mechanism.

5. Mechanism: the gate does not gate by context

Three probes, each applied to fully trained networks. The headline numbers below are MNIST seed 0; the pattern replicates on Fashion-MNIST seed 0 and MNIST seed 1 (all numbers in the artifact): at b=0, flip rates

14.8–30.1% mean / 23.8–51.7% worst pair; at b=10, flip rates 0.007–0.021% mean / at most 0.15% worst pair; gate own/other identical to four decimals (0.564–0.565); and mean column cosines in $[0.001, 0.003]$ in every case.

Prediction flip rate.. For each task c and each alternative context $c' \neq c$, measure the fraction of test samples whose argmax prediction differs between ctx=c and ctx=c'. Base network: mean 14.8% (MNIST) and 30.1% (Fashion-MNIST), worst pair up to 51.7% — the base output is genuinely, substantially context-sensitive at the individual-prediction level, yet none of that sensitivity survives to final accuracy (Sec. 4: at most 0.25 points). The context pathway does something per sample; what it does nets out to nothing. Repaired network (b=10): mean 0.007–0.021%, worst pair 0.15% — no longer *exactly* zero, but a rounding error: roughly one flipped prediction in five thousand on average, a dozen samples for the worst single pair.

Gate response.. The winning segment's gate value $\sigma(d)$ has mean 0.565 under the correct context and 0.565 under every alternative — identical to four decimal places, in both the base (0.5651) and repaired (0.5652) networks. The context one-hot selects *which* segment wins, but not *how much* it gates: the gate's value is a constant of the training run, not a function of the context.

Column geometry.. The dendritic weights projected onto the T context columns remain mutually orthogonal after training: mean pairwise cosine 0.003 (per-model mean in $[0.001, 0.003]$; the most negative single pair over the three models is -0.056), indistinguishable from their random initialization. Training, with per-task gradient protection, leaves the context channel of the gate bank essentially at its prior.

Together: the architecture's context pathway is present, trainable in principle, protected per task — and unused as a *modulator*. The max over segments is won by whichever row the (orthogonal, near-random) columns happen to favor, the gate value is the same constant (≈ 0.565) no matter which segment wins, and the repair adds a constant so large (b=10 against $|u| \lesssim 1$) that the own segment always wins with $\sigma(10+u) \approx 1$: a context-free gain of one. Equation 1 degenerates, in both arms, to a multiplication by a constant — ≈0.57 in the base network, ≈1.0 in the repaired one. Which of those two constants matters, and why, is what the next section measures.

6. Ablation: what does the repair repair?

If the repaired network computes (approximately) $h = \text{kwta}(t)$, then a network with the dendritic bank deleted and the same trunk, sparsifier, and anchor chain should reproduce most of the repair's benefit — and, conversely, a network whose gate is held at the base system's measured constant (0.565) should reproduce the base system, because the base gate already *is* that constant (Sec. 5). We therefore run four arms under one protocol (ρ=0, n=10 paired seeds, all arms at n_h=256, all with usage-weighted Fisher anchors at λ=1000): **base** (b=0), **repaired** (b=10), **no-gate** ($h = \text{kwta}(W_{\text{ff}}x)$; the bank deleted), and **constant-gate** ($h = \text{kwta}(0.565 \cdot W_{\text{ff}}x)$; the gate pinned at the base system's measured mean). The base and repaired arms replicate the Sec. 4 runs to two decimals. The degenerate controls were added during the post-audit re-run (Sec. 2); they were not preregistered. The no-gate control exists in three versions in the artifact — an invalidated first attempt (an extra sigmoid on t that changed the activation structure), a width-mismatched

Table 2: Four-arm control experiment (n=10 paired seeds, ρ=0, matched width n_h=256 and matched usage-weighted anchors). Bold marks the best arm per dataset.

dataset	base (b=0)	const-gate (0.565)	repaired (b=10)	no-gate
MNIST	63.86	63.62	**68.67**	67.50
Fashion	42.20	42.70	42.89	**44.27**

second version (n_h=2000), and the matched version reported here — all retained under an explicit deprecation notice.

The constant-gate control reproduces the base system, not the repair. Pinning the gate at 0.565 lands at 63.62 on MNIST (base: 63.86; paired difference -0.25, $t = -0.7$, n.s.) and 42.70 on Fashion (base: 42.20; $+0.49$, t=0.7, n.s.). Holding the gate constant buys nothing — which is exactly what the mechanism section predicts, since the base network's gate already *is* constant at that value. The entire difference between the base and repaired networks is the value of that constant: 0.57 damps the trunk, 1.0 does not.

Gate deletion recovers most of the repair — and on Fashion-MNIST, more than the repair. The no-gate control reaches 67.50 on MNIST, recovering +3.64 of the repair's +4.80-point gain (76%) and falling short of the repaired network by -1.16 ($t = -2.7$): keeping the (saturated) multiplicative structure is worth about one point, marginally significant at n=10. On Fashion-MNIST the no-gate control is the *best* arm: +2.07 over base (t=2.5) and +1.37 over the repaired network (t=1.9, n.s.) — on the harder dataset, full deletion beats the repair outright.

Sensitivity to anchor weighting. A second no-gate run with plain

(unweighted) Fisher anchors — a stronger, mismatched chain — lands at 63.63 on MNIST and 41.10 on Fashion, i.e. at or below base. Anchor weighting alone moves the no-gate arm by 3.9 and 3.2 points. The matched-anchor four-arm table above is therefore the primary comparison, and the residual MNIST gap (-1.16) should be read against that sensitivity. The first version of this experiment reported a clean MNIST *equivalence* between the no-gate control and the repair; that verdict was an artifact of the width mismatch disclosed in Sec. 2 and is retracted.

Verdict. The repair is a saturating soft deletion: b=10 drives $\sigma \to 1$, which is the no-gate limit reached through the gate rather than around it. What the repair buys on these benchmarks is the removal of a constant $\approx$0.57 damping factor — and on Fashion-MNIST, where deletion is free to remove it *exactly*, the removal is worth slightly more than the repair itself. What does *not* follow is the stronger claim of the first version, that the repair and gate removal are interchangeable on MNIST: with matched widths and matched anchors they are close (76% of the gain recovered) but distinguishable.

7. Discussion

For the S1 story.. The dose–response audit established that the own-segment bias repairs the learning term; the mechanism section of that paper showed the network never pushes back at the parameter level. This paper supplies the missing re-qualification: the repair does not restore routing — there was no routing to restore — it *saturates the gate into a context-free gain of one*, i.e. a soft deletion of a constant $\approx$0.57 damping factor. Read together, the chain is now: no specialization (S1) $\to$ repair works (S1) $\to$ the repair

is a saturating gate deletion that recovers 76% of the removal benefit on MNIST and is beaten by full removal on Fashion-MNIST (this paper) → the repair does not transfer to strong backbones (S3). What survives as the "mechanism" of the repair is the removal of a constant damping — sparse capacity management at best, context-gated routing not at all.

For evaluation practice.. Robustness certificates obtained under context corruption are usually read as good news. In this family they are a diagnostic: a system whose accuracy is invariant to *adversarial* context corruption is a system whose context pathway is not load-bearing. The three-instrument protocol (Sec. 4–6) costs a few GPU-hours and converts that ambiguity into a verdict. We suggest it as a standing companion to any context-gated continual-learning evaluation, in the same spirit as the dead-neuron probe of S1.

Limitations.. (i) Noise is injected at evaluation only; training-time context errors corrupt usage statistics and anchors differently, and are the natural next experiment. (ii) The mechanism probes are computed on fully trained networks at T=10; whether the context columns stay orthogonal at larger T or longer training is untested. (iii) The degenerate controls were not pre-registered and were added after the audit; the anchor-weighting sensitivity of the no-gate arm (±3.9 points, Sec. 6) is larger than the residual repair-vs-deletion gap it bounds. (iv) One architecture family, two digit datasets — the same scope limits as the companion line.

8. Conclusion

Asked whether context-gated protection survives a wrong context signal, the system answered: it never needed the signal. The repair that was motivated by broken routing survives for the same reason — it never used the routing; it removes a constant damping factor that the never-used gate had left in place, and where removal can be done exactly, it does slightly better than the repair. The general lesson travels beyond this architecture: *when a robustness probe returns exactly zero damage, measure the pathway before celebrating the armor — and when a repair works, check whether it works by fixing the mechanism or by deleting it.*

Data and code availability

All code, per-run records (including per-ρ accuracy vectors), the preregistration document, and both versions of the degenerate control (including the invalidated first attempt) are in the artifact repository https://github.com/zhouranxi/delp-anchor-budget (folder `S4_gate_bypass/`, self-contained: paper sources, scripts, raw records), alongside the companion papers S1–S3.

Declaration of competing interest

The author declares no competing interests.

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